

Techy Campus NMEA Location Reference

This document converts the latitude and longitude fields in the employee NMEA GGA logs into named Techy campus locations. All coordinates and locations are synthetic.

How to Read the GGA Records

Each log line contains a local timestamp followed by a standard GGA sentence. For this test, the important GGA fields are UTC time, latitude, N/S, longitude, and E/W. Example structure:
\$GPGGA,UTC,Latitude,N,Longitude,W,FixQuality,Satellites,HDOP,Altitude,M,...*Checksum.

The latitude field uses DDMM.MMMM format and the longitude field uses DDDMM.MMMM format. You do not need to convert them to decimal degrees: compare the numeric NMEA latitude and longitude fields directly with the geofence ranges below.

Campus Geofences

Location	NMEA latitude field	NMEA longitude field	Type
E101 - Ethan Cole desk	3252.8065-3252.8097 N	11714.0884-11714.0923 W	Assigned desk
E102 - Maya Shah desk	3252.8065-3252.8097 N	11714.0820-11714.0859 W	Assigned desk
E103 - Jordan Lee desk	3252.8065-3252.8097 N	11714.0756-11714.0794 W	Assigned desk
Meeting Room Atlas	3252.8065-3252.8097 N	11714.0627-11714.0678 W	Meeting room
Meeting Room Beacon	3252.8130-3252.8162 N	11714.0627-11714.0678 W	Meeting room
Campus Gym	3252.8281-3252.8367 N	11714.0820-11714.0923 W	Gym
Cafeteria	3252.8281-3252.8367 N	11714.0627-11714.0730 W	Cafeteria
Transit / Hallway	3252.8184-3252.8248 N	11714.0704-11714.0781 W	Transit

Classification Rules

- A GGA record belongs to a named location only when both its NMEA latitude field and NMEA longitude field fall inside that location's ranges.
- The N/S and E/W hemisphere indicators must also match the reference. All Techy test locations use N and W.
- If a record does not fall inside any named geofence, classify it as Unknown unless another document explicitly defines the location.
- The generated GGA points include small GPS-like variation but remain inside the intended geofence.
- Classify every 10-minute record independently. Do not infer a location only from the employee's previous or next record.